

Loss Surface Analysis of the Hybrid Model

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Abstract

This report investigates the local geometry of the loss surface around fitted parameter values of the Hybrid behavioural model. For each fitted solution, random one-dimensional slices through the seven-dimensional parameter space were sampled and the loss was recomputed along each direction. Surprisingly, even with a small number of random directions, several lower-loss parameter settings were found near the fitted solutions. This suggests that the objective landscape may be non-smooth, weakly identifiable, plateau-like, or contain saddle-like regions, which can make gradient-free optimisation methods converge to solutions that are not locally optimal under this diagnostic.

Keywords

Hybrid model, loss surface, parameter optimisation, TPE, CMA-ES, behavioural modelling

1. Methods

The aim of this analysis was to probe the local loss surface around fitted parameter values of the Hybrid model. The model was fitted separately for each subject and optimisation method, and the fitted parameter vector was denoted by

$$\theta^* = [\sigma_{\text{noise}}, A_{\text{repulsion}}, \gamma, \sigma_{\text{update}}, \eta_{\text{learning}}, \sigma_{\text{boundary}}, \alpha]^T \in \mathbb{R}^7. \quad (1)$$

The seven fitted parameters were constrained to the following ranges:

$$\sigma_{\text{noise}} \in [0.01, 0.45], \quad (2)$$

$$A_{\text{repulsion}} \in [0, 0.5], \quad (3)$$

$$\gamma \in [0.01, 1], \quad (4)$$

$$\sigma_{\text{update}} \in [0.01, 0.7], \quad (5)$$

$$\eta_{\text{learning}} \in [0.01, 1], \quad (6)$$

$$\sigma_{\text{boundary}} \in [0.01, 10], \quad (7)$$

$$\alpha \in [0, 1]. \quad (8)$$

For a given parameter vector θ , the loss was defined as the discrepancy between the model-generated matrix and the empirical target matrix:

$$\mathcal{L}(\theta) = E(M_{\text{model}}(\theta), M_{\text{data}}), \quad (9)$$

where $E(\cdot, \cdot)$ is the matrix error function used during fitting. In the present analysis, the conditional matrix was used as the target matrix:

$$M_{\text{data}} = M_{\text{conditional, data}}. \quad (10)$$

To investigate the local loss surface, random directions were sampled in the seven-dimensional parameter space. A raw random vector was first drawn from an isotropic Gaussian distribution:

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I_7). \quad (11)$$

Because the seven parameters have very different numerical ranges, the random vector was scaled by the width of each parameter range:

$$r_i = \theta_{i, \max} - \theta_{i, \min}. \quad (12)$$

The scaled random direction was then normalised:

$$\mathbf{v} = \frac{\mathbf{z} \odot \mathbf{r}}{\|\mathbf{z} \odot \mathbf{r}\|_2}, \quad (13)$$

where $\odot$ denotes element-wise multiplication. This scaling makes the random direction more meaningful, because otherwise parameters with larger numerical ranges, such as σ_{boundary} , could dominate the geometry of the sampled directions.

For each random direction $\mathbf{v}$, a one-dimensional slice through the fitted point was evaluated:

$$\boldsymbol{\theta}(\lambda) = \text{clip}(\boldsymbol{\theta}^* + \lambda \mathbf{v}), \quad (14)$$

where $\text{clip}(\cdot)$ enforces the parameter bounds. The scalar displacement parameter λ was sampled over

$$\lambda \in [-1, 1], \quad (15)$$

using 41 evenly spaced values:

$$\Delta\lambda = \frac{2}{40} = 0.05. \quad (16)$$

For each subject and optimisation method, $D = 10$ random directions were sampled. Therefore, each fitted solution was tested at

$$10 \times 41 = 410 \quad (17)$$

nearby parameter settings.

2. Analysis

2.1. Subject 6269757 fitted with CMA-ES

For subject 6269757 fitted using CMA-ES, the fitted loss was

$$\mathcal{L}(\boldsymbol{\theta}^*) = 0.055159. \quad (18)$$

The best point found by the local random-direction search had loss

$$\mathcal{L}_{\text{best}} = 0.053293. \quad (19)$$

This corresponds to a relative improvement of approximately

$$\frac{0.055159 - 0.053293}{0.055159} \times 100 \approx 3.38\%. \quad (20)$$

The fitted parameters were

$$\begin{aligned} \sigma_{\text{noise}} &= 0.118160, & A_{\text{repulsion}} &= 0.264088, \\ \gamma &= 0.842005, & \sigma_{\text{update}} &= 0.486071, \\ \eta_{\text{learning}} &= 0.336001, & \sigma_{\text{boundary}} &= 4.726866, \\ \alpha &= 0.693673. \end{aligned} \quad (21)$$

The best point was found in direction 1 at $\lambda = 0.15$, with Euclidean distance 0.15 from the fitted parameter vector. The best found parameters were

$$\begin{aligned} \sigma_{\text{noise}} &= 0.123349, & A_{\text{repulsion}} &= 0.267788, \\ \gamma &= 0.847915, & \sigma_{\text{update}} &= 0.486469, \\ \eta_{\text{learning}} &= 0.346986, & \sigma_{\text{boundary}} &= 4.577558, \\ \alpha &= 0.690367. \end{aligned} \quad (22)$$

Although the improvement is modest, the fact that a lower-loss point was found with only 10 random directions suggests that the CMA-ES fitted point is not perfectly locally optimal under this diagnostic. The zoomed plot shows that the lower-loss point is close to the fitted solution, indicating that small parameter changes can improve the loss.

2.2. Subject 6269757 fitted with TPE

For the same subject fitted with TPE, the fitted loss was

$$\mathcal{L}(\theta^*) = 0.057932. \quad (23)$$

The best point found by the random-direction search had loss

$$\mathcal{L}_{\text{best}} = 0.051438. \quad (24)$$

This corresponds to a relative improvement of approximately

$$\frac{0.057932 - 0.051438}{0.057932} \times 100 \approx 11.21\%. \quad (25)$$

The fitted parameters were

$$\begin{aligned} \sigma_{\text{noise}} &= 0.126412, & A_{\text{repulsion}} &= 0.339304, \\ \gamma &= 0.939783, & \sigma_{\text{update}} &= 0.272976, \\ \eta_{\text{learning}} &= 0.547309, & \sigma_{\text{boundary}} &= 9.295416, \\ \alpha &= 0.451329. \end{aligned} \quad (26)$$

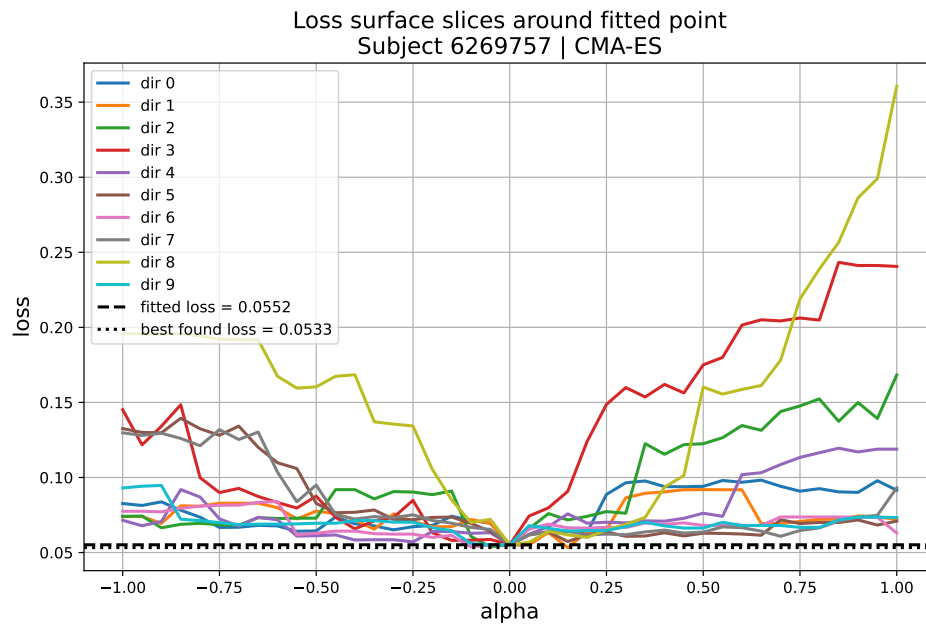


Figure 1: Loss surface slices around the fitted point for subject 6269757 fitted with CMA-ES. Each coloured curve corresponds to one random direction in the seven-dimensional parameter space. The dashed black line shows the fitted loss and the dotted black line shows the best loss found along the sampled directions.

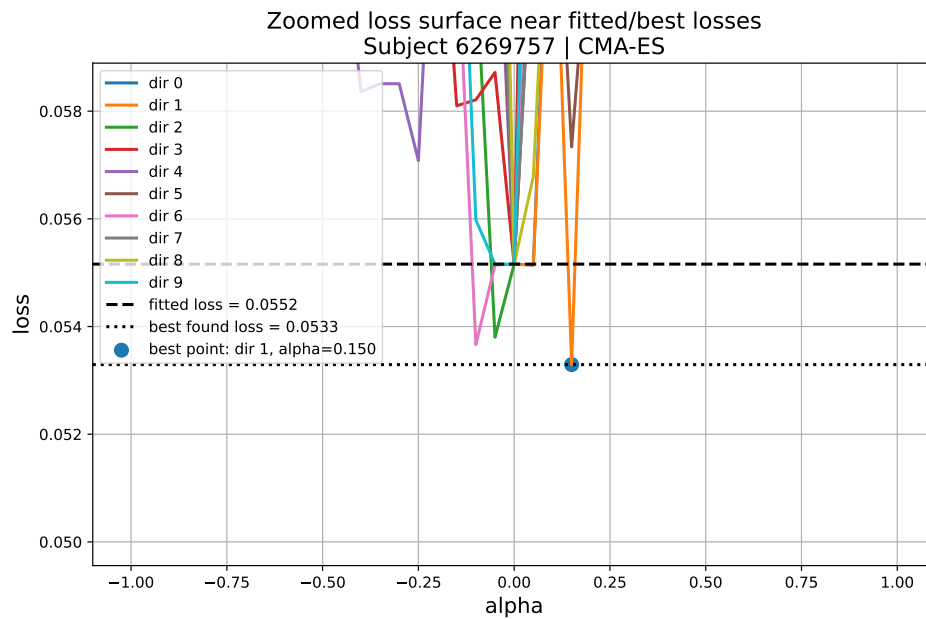


Figure 2: Zoomed view of the loss surface for subject 6269757 fitted with CMA-ES. The best lower-loss point was found in direction 1 at $\lambda = 0.15$.

The best point was found in direction 4 at $\lambda = -0.20$, with Euclidean distance 0.20 from the fitted parameter vector. The best found parameters were

$$\begin{aligned}\sigma_{\text{noise}} &= 0.125497, & A_{\text{repulsion}} &= 0.338246, \\ \gamma &= 0.919371, & \sigma_{\text{update}} &= 0.280447, \\ \eta_{\text{learning}} &= 0.550948, & \sigma_{\text{boundary}} &= 9.096739, \\ \alpha &= 0.445032.\end{aligned}\tag{27}$$

The improvement for TPE is larger than for CMA-ES in this subject. This suggests that the TPE solution may be located on a less stable or less locally optimal region of the loss surface. The sampled curves also show substantial variability across directions, consistent with an irregular objective landscape.

2.3. Subject 6269770 fitted with CMA-ES

For subject 6269770 fitted using CMA-ES, the fitted loss was

$$\mathcal{L}(\theta^*) = 0.041005.\tag{28}$$

The best point found by the local random-direction search had loss

$$\mathcal{L}_{\text{best}} = 0.031493.\tag{29}$$

This corresponds to a relative improvement of approximately

$$\frac{0.041005 - 0.031493}{0.041005} \times 100 \approx 23.20\%.\tag{30}$$

The fitted parameters were

$$\begin{aligned}\sigma_{\text{noise}} &= 0.104845, & A_{\text{repulsion}} &= 0.459508, \\ \gamma &= 0.770726, & \sigma_{\text{update}} &= 0.398532, \\ \eta_{\text{learning}} &= 0.688318, & \sigma_{\text{boundary}} &= 0.388483, \\ \alpha &= 0.979646.\end{aligned}\tag{31}$$

The best point was found in direction 0 at $\lambda = 0.50$, with Euclidean distance 0.499729 from the fitted parameter vector. The best found parameters were

$$\begin{aligned}\sigma_{\text{noise}} &= 0.121024, & A_{\text{repulsion}} &= 0.500000, \\ \gamma &= 0.805531, & \sigma_{\text{update}} &= 0.302862, \\ \eta_{\text{learning}} &= 0.783682, & \sigma_{\text{boundary}} &= 0.862936, \\ \alpha &= 0.922516.\end{aligned}\tag{32}$$

This case shows the largest improvement among the examples reported here. The improvement is especially notable because it was found by a simple random-direction diagnostic rather than a dedicated second-stage optimiser. The best point also reached the upper bound of $A_{\text{repulsion}}$, suggesting that the fitted solution may be constrained by the parameter range or that the optimum may lie near the boundary.

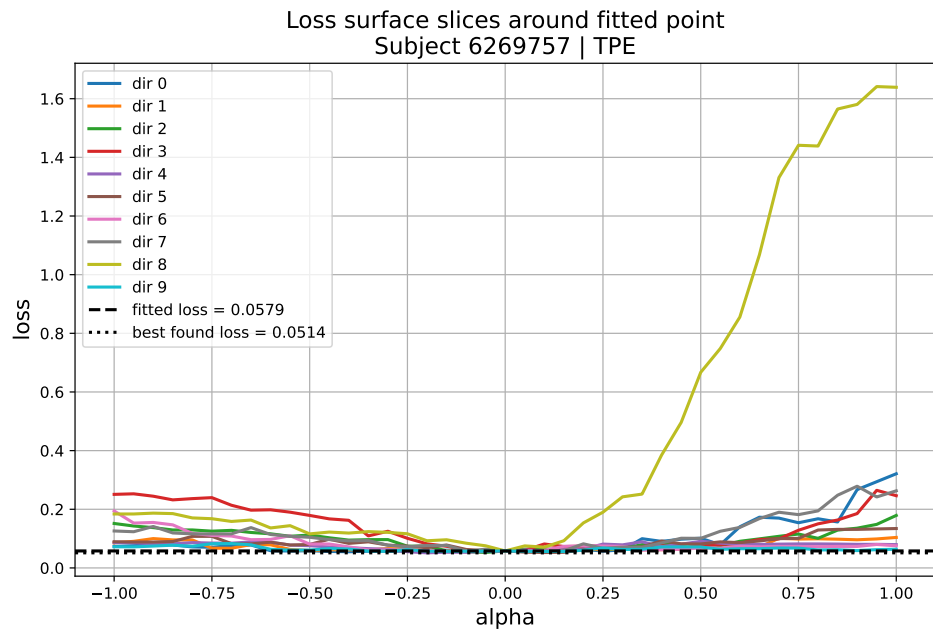


Figure 3: Loss surface slices around the fitted point for subject 6269757 fitted with TPE. The loss curves show stronger variation than in the CMA-ES fit, and a lower-loss point was found within the sampled directions.

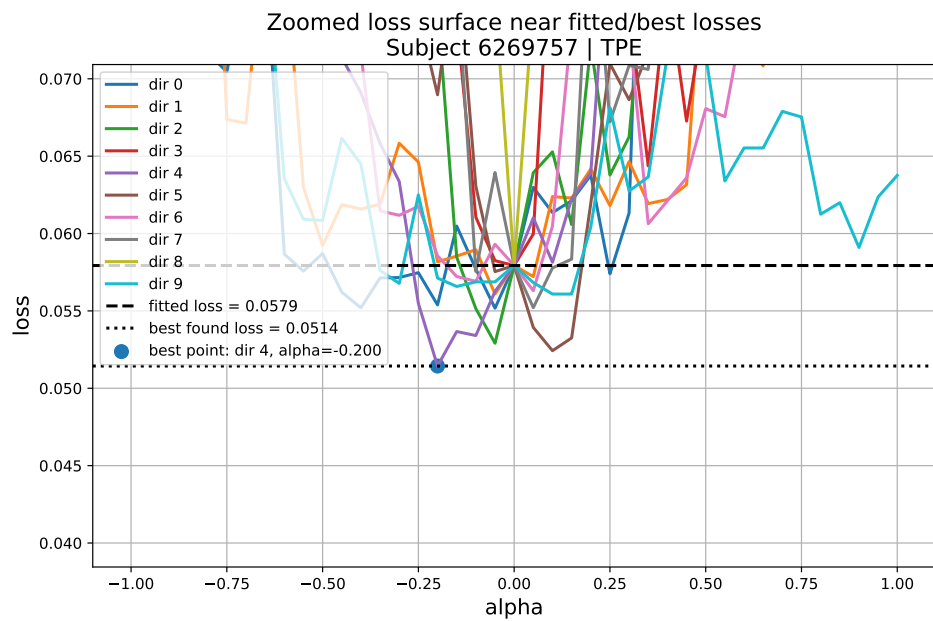


Figure 4: Zoomed view of the TPE loss surface for subject 6269757. The best lower-loss point was found in direction 4 at $\lambda = -0.20$.

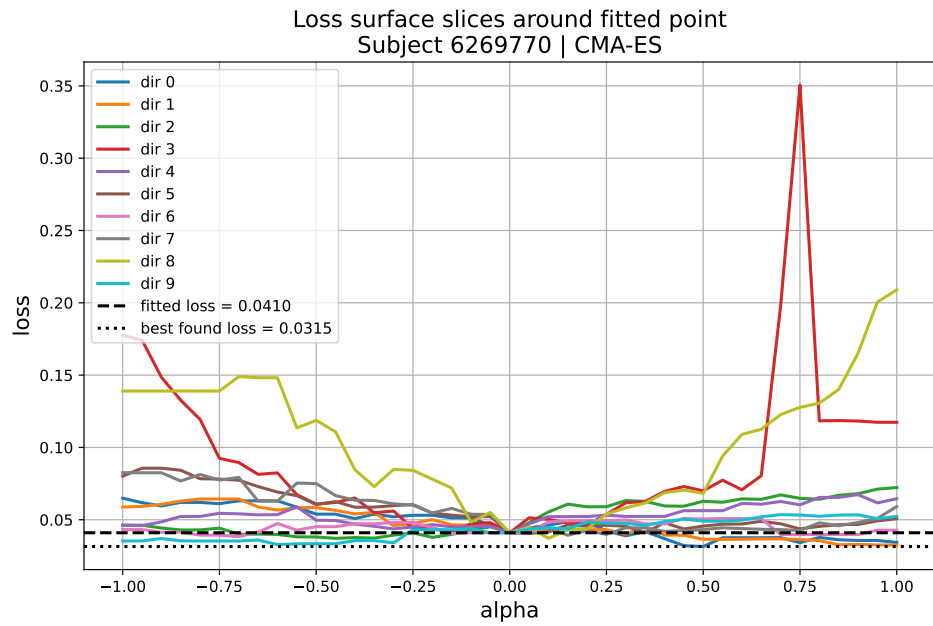


Figure 5: Loss surface slices around the fitted point for subject 6269770 fitted with CMA-ES. A substantially lower-loss point was found along direction 0.

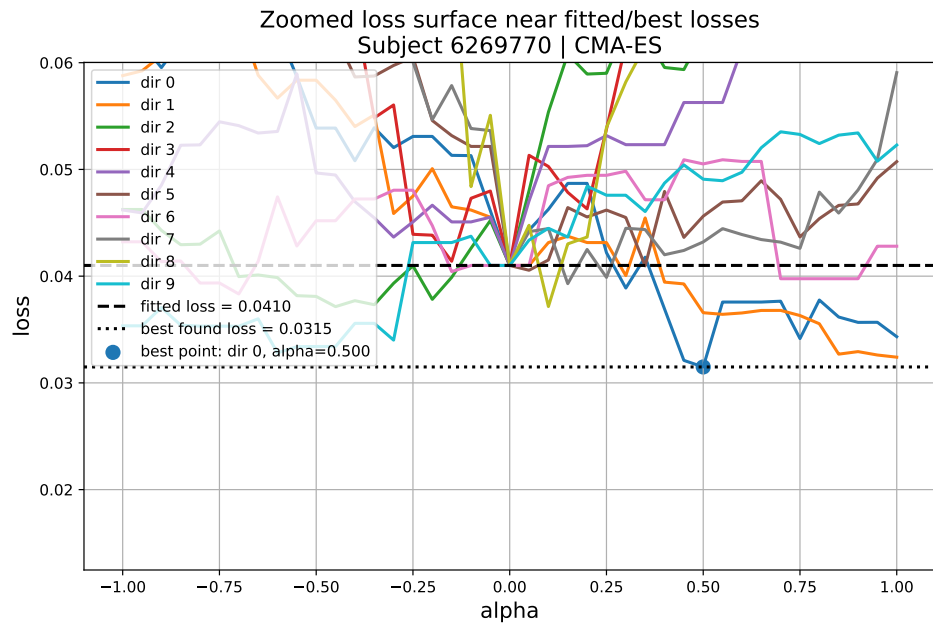


Figure 6: Zoomed view of the CMA-ES loss surface for subject 6269770. The best point was found at $\lambda = 0.50$, suggesting that the fitted point may lie on a slope or near a broad non-optimal region.

2.4. Subject 6269770 fitted with TPE

For subject 6269770 fitted using TPE, the uploaded loss-surface figure shows a fitted loss of approximately

$$\mathcal{L}(\theta^*) \approx 0.0276, \quad (33)$$

and a best found loss of approximately

$$\mathcal{L}_{\text{best}} \approx 0.0254. \quad (34)$$

This corresponds to an approximate relative improvement of

$$\frac{0.0276 - 0.0254}{0.0276} \times 100 \approx 7.97\%. \quad (35)$$

The best point was found in direction 8 at $\lambda = 0.10$ according to the zoomed figure. Full parameter values for this case were not available in the numerical printout included in this report.

2.5. Summary of numerical results

2.6. Interpretation

Across these examples, lower-loss points were found despite sampling only 10 random directions around each fitted solution. This was not expected, because if the optimiser had converged to a strong local optimum, most sufficiently small perturbations around the fitted point would be expected to increase the loss. Instead, the sampled slices frequently revealed directions along which the loss decreased.

There are several possible explanations. First, the loss surface may be highly non-smooth due to stochastic simulation, finite trial effects, matrix estimation noise, and clipping of parameters at their bounds. Second, the fitted solution may lie near a saddle-like region, where the loss increases in some directions but decreases in others. Third, the surface may contain broad plateaus, where many parameter combinations give similar losses and small random movements can occasionally lead to better values. Fourth, some parameters may be weakly identifiable, meaning that different parameter combinations can generate similar behavioural matrices.

The results also show that different optimisation methods can converge to different regions of parameter space for the same subject. For subject 6269757, the CMA-ES and TPE fitted solutions had different fitted losses and different local geometries. For subject 6269770, the TPE fitted loss was lower than the CMA-ES fitted loss, but even the TPE solution still had a nearby lower-loss point in the sampled directions. This suggests that the issue is not specific to one optimiser, but may reflect the underlying geometry of the Hybrid-model objective function.

Overall, this diagnostic suggests that the fitted solutions should be interpreted with caution. The presence of nearby lower-loss points indicates that the optimisation landscape may contain local irregularities, ridges, plateaus, or saddle-like structures. A useful next step would be to use these lower-loss points as new initialisations for an additional optimisation stage, or to perform repeated local searches around the fitted solutions to better characterise the nearby geometry of the loss surface.

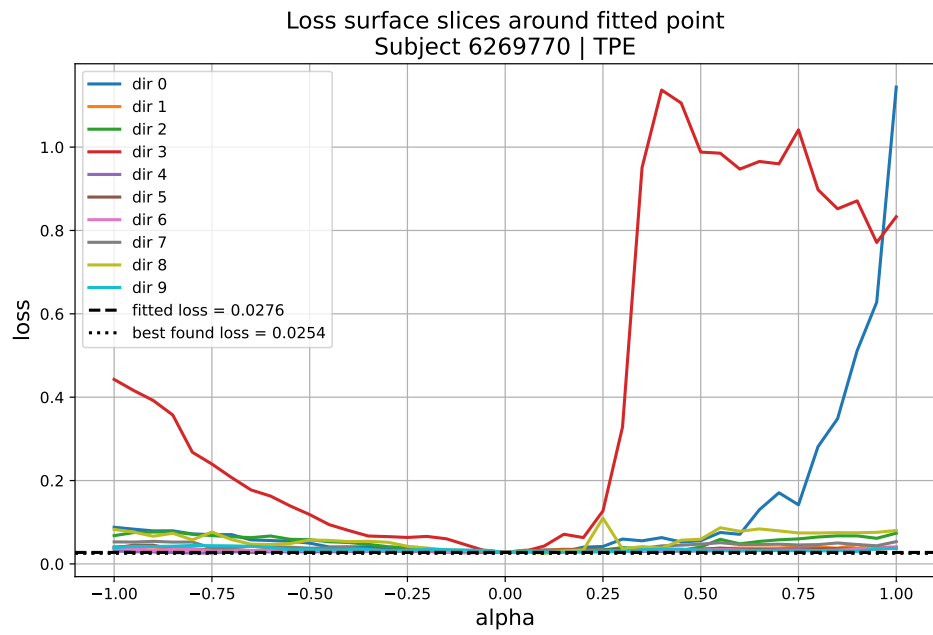


Figure 7: Loss surface slices around the fitted point for subject 6269770 fitted with TPE. Even though the fitted loss is lower than the CMA-ES result for this subject, the random-direction analysis still found a nearby lower-loss point.

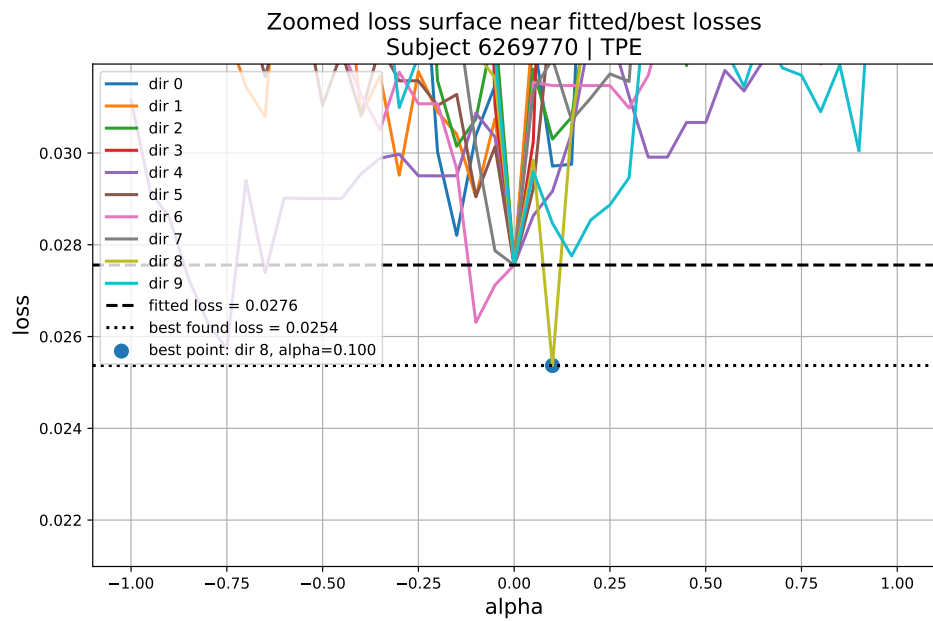


Figure 8: Zoomed view of the TPE loss surface for subject 6269770. The best point was found in direction 8 at $\lambda = 0.10$.

Table 1

Summary of fitted and best-found losses from the local random-direction loss-surface analysis.

Subject	Optimiser	Fitted loss	Best found loss	Best direction	Best λ
6269757	CMA-ES	0.055159	0.053293	1	0.150
6269757	TPE	0.057932	0.051438	4	-0.200
6269770	CMA-ES	0.041005	0.031493	0	0.500
6269770	TPE	≈ 0.0276	≈ 0.0254	8	0.100